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II. Causational Inference with the NEMSIS Dataset

Much of the emergency response system is based on the founding pillar of “faster response, better outcomes.” But is that true? This section aims to answer a version of this question: within rural communities, how are patient outcomes affected by response times above the national average?

Abstract

Using 2021 NEMSIS data, a causal analysis of 911-initiated ground responses in rural areas was conducted to estimate the effect of delayed EMS response on clinical deterioration, defined as a worsening of acuity between EMS arrival and patient handoff. Propensity score matching was used, along with a sufficient adjustment set, including response delay, bystander assistance, and initial patient acuity. Logistic regression models across multiple delay thresholds revealed no significant effect at 5 minutes, but significant and increasing effects at 9 (OR = 1.25), 11 (OR = 1.35), and 20 minutes (OR = 1.65). Bystander intervention was strongly protective in all models. These findings support evidence-based response benchmarks and underscore the importance of both minimizing delay and enhancing pre-arrival community intervention.

Background

In rural America, 1 in 10 callers will wait over 30 minutes for an ambulance to arrive. Rural patients experiencing a medical emergency face a double challenge: they often are farther from EMS responders and definitive care facilities. The travel distances in rural areas, sometimes exacerbated by difficult terrain or limited road networks, directly translate into delayed treatment. Multiple studies show that EMS response times are significantly longer in rural areas compared to urban settings. In a recent systematic review of 37 studies, 93% of the comparisons found that rural EMS response times were significantly longer than urban response times.¹ National EMS data indicate a median response time of roughly 7 minutes from 911 call to first unit arrival overall, but that median climbs to over 14 minutes in rural settings, with nearly 1 in 10 rural emergencies waiting about 30 minutes for EMS to arrive. In contrast, the median response in dense urban areas is around 6–7 minutes. Analysis of over 1.7 million EMS encounters found that 90th-percentile response times were ~12 minutes in urban zip codes vs. ~30 minutes in rural zip codes.² Taken together, the evidence paints a clear picture of disparity: rurality is strongly associated with slower EMS response, which plausibly contributes to poorer patient outcomes. The key question, however, is

¹ [Prehospital Time Interval for Urban and Rural Emergency Medical Services: A Systematic Literature Review](#)

² [Emergency Medical Services Response Times in Rural, Suburban, and Urban Areas - PMC](#)

whether these delays necessarily cause worse outcomes or merely correlate with other factors that put patients at risk.

Within urban environments, some research has been conducted. An urban study analyzed over 5,400 high-priority EMS responses and found no significant survival difference once response times were beyond 5 minutes.³ Patients reached within 5 minutes had a mortality of 0.51% versus 1.58% for those who waited longer than 5 minutes. The authors cautioned that variables other than time (e.g. the nature of calls or patient characteristics) may have influenced the better outcomes in the sub-5-minute group. Another study in an urban EMS system examined trauma patients with response times under versus over 8 minutes and found no overall survival difference between the groups, after accounting for injury severity. Intriguingly, the >8-minute group had slightly better survival for the most critically injured patients, possibly reflecting that some very severe trauma victims died quickly on scene (within minutes, before EMS arrival) regardless of EMS speed, whereas those who survived long enough for a slower response were a biased subset. Noticeably, there is a gap in the literature regarding the effect of response times and patient outcomes in the rural setting. We therefore conduct analysis on the rural community and seek to understand if a delay in response times leads to worse patient outcomes.

Methods

The NEMSIS Dataset

This analysis utilizes data from the National Emergency Medical Information System (NEMSIS), a comprehensive repository of EMS activation and patient care reports (PCRs) in the United States. NEMSIS maintains and outputs this yearly dataset to inform providers, educators, and researchers on the efficacy and state of modern EMS. The information is voluntarily submitted by State and Territory EMS Officials, and not all agencies or every EMS activation is necessarily represented.

The dataset includes millions of lines, each representing a de-identified EMS activation and subsequent patient care report (PCR). Each PCR details every aspect of an EMS call: from dispatch, to patient hand off, and everything in between. This includes ambulance response mode, patient demographics, patient outcome, and many more.

We use data from 2021, focusing specifically on 911-initiated ground responses in rural settings, excluding interfacility transports, standbys, and cases with non-transport outcomes (e.g., patient refusals, crew cancels, or no patient found). Only PCRs in which care and transport were rendered by the initiating EMS personnel were retained. Additionally, to reduce confounding from regional variability in dispatch behavior and

³ [comparison of response time and survival in an urban emergency medical services system - PubMed](#)

geography, we restricted analysis to cases explicitly marked as “rural” based on the urbanicity variable.

Dataset Processing and Variable Construction

The raw dataset comprised tens of millions of observations and required substantial preprocessing due to size (~375 GB) and re-encoding. All processing was conducted on a high-memory workstation. Data cleaning involved standardizing inconsistent formats, imputing or releveling certain NA values, where justifiable, and excluding fields with unresolvable ambiguity. For example, for the variable `response_mode_descriptors`, “lights and sirens” outnumbered “no lights or sirens” threefold, which does not intuitively make sense. However, NA values are reasonably the same as no response mode descriptors, thus they are coded as “no lights or sirens.” This methodology could not be applied to all variables, as many did not have an intuitive default. Variables such as response time, transport time, bystander presence, and delay status were binarized to facilitate causal analysis.

Our primary outcome variable is clinical deterioration, constructed by comparing initial and final patient acuity. Acuity was categorized according to NEMSIS designations: Lower Acuity (Green), Emergent (Yellow), Critical (Red), and Dead (Black). We defined “worsened” (1) as a transition to a more severe acuity or death, and “unchanged/improved” (0) as no change or movement toward less severity. This was encoded as a binary variable `pt_worsened`. A number of other variables were dichotomized for binary analysis.

The treatment variable `above_med_response_time` was constructed to indicate whether the EMS response time exceeded a specified threshold. We tested this treatment definition at multiple thresholds (5, 9, 11, and 20 minutes) to examine dose–response relationships. Thresholds were selected based on national benchmarks and prior literature, with the 9-minute and 11-minutes cutoff serving as the primary reference as it was the 2021 national median and average response time, respectively, for rural EMS.

Covariates included binary indicators for use of lights and sirens, presence of any response delay, and bystander aid rendered before EMS arrival. We also controlled for patient age to assess robustness to demographic confounding.

The resulting dataset we conduct our analysis on is about 10gb in size, containing TKTKT rows and columns.

Exploratory Data Analysis

not yet complete

- times

- location types
- summaries on the confounders and treatment and outcome
- pt ages
- call times, day of week etc
- basically everything in the prev, and CI rmd's

Fundamental Problem of Causal Inference

The fundamental problem states that one can never observe both potential outcomes for the same case simultaneously. Here, we cannot know the outcome of an identical call with different response times as calls are not repeatable. That is, for a given EMS activation, we can only observe the outcome for either the faster response time or the slower response time, but not both.

Treatment and Outcome

Treatment A is the binary variable that indicates whether the EMS response time was at or above the cutoff time.

$A_0 = \text{response time below the cutoff}$

$A_1 = \text{response time at or above the cutoff}$

Outcome Y is the binary variable representing patient condition throughout the call.

$Y_0 = \text{patient condition unchanged}$

$Y_1 = \text{patient condition worsens}$

Directed Acyclic Graph

To guide our causal identification strategy, we constructed a directed acyclic graph (DAG) representing the hypothesized relationships between EMS response time, patient deterioration, and relevant covariates. The DAG includes the following variables:

D - Response delay. (e.g. traffic, weather, wrong directions, scene access, etc).

R - Response type. (lights and sirens or no lights and sirens).

I - Initial patient acuity (Dead/Black, Critical/Red, Emergent/Yellow, or Lower Acuity/Green)

C - Patient/caller complaint. (This often dictates response type).

B - Were bystanders on scene to help/provide immediate care before EMS arrival?

Y: Outcome (patient deterioration).

A: Treatment assignment (response time above threshold).

We assume that response delay (D), initial acuity (I), and bystander help (B) all influence both response time (A) and patient deterioration (Y), making them confounders. An early bystander (B), for example, likely will be able to give more context to dispatch and start rendering aid. And initial acuity (I) is what chiefly dictates response type. Initial acuity is formed from a number of factors, but isn't always necessarily accurate: the caller could be wrong, or the dispatch could mis-interpret. Response delays (R) are their own independent factor in response time.

Response mode (R) is treated as a downstream proxy of perceived acuity that influences A but does not directly affect Y. This avoids collider bias that would arise from adjusting for a mediator.

need to insert dag here

Thus, the DAG reveals the following non-causal paths from A to Y:

$$A \leftarrow D \rightarrow Y$$

$$A \leftarrow I \rightarrow Y$$

$$A \leftarrow B \rightarrow Y$$

To block these paths and satisfy conditional exchangeability, we adjust for the following sufficient set of pre-treatment covariates:

$$\{D, I, B\}$$

All adjustment variables are measured prior to treatment assignment and are not affected by post-treatment mechanisms, preserving the causal interpretation of the treatment effect.

Plausibility for Causal Inference

While we can never prove true causal inference, especially in an observational dataset, we may become more persuasive in an argument for our causal inference analysis by ensuring conditional exchangeability, positivity, and consistency.

Conditional Exchangeability

The assumption of conditional exchangeability requires that, on a set of observed covariates, the potential outcomes are independent of treatment. In this setting, this implies that after adjusting for relevant pre-treatment variables, such as lights and

sirens use or bystander intervention, the assignment of EMS response time (above or below the threshold) conditional exchangeability holds. This assumption is valid through our use of propensity score matching to balance covariates and restrict comparison to treated and control cases with identical or similar characteristics.

However, it is acknowledged that the potential for residual confounding factors exist, particularly from patient-level clinical variables that are either poorly measured or unavailable in the NEMSIS dataset. These could include comorbidities, preexisting health status, or previous frequent use of emergency response systems. These factors could jointly affect EMS response and patient outcome. Although we mitigate this by excluding non-transport and non-emergent calls, and by adjusting for available covariates like age in sensitivity models, conditional exchangeability remains a strong assumption that may not fully hold. As such, with all causal analyses, findings may only ever be suggestive of a causal relationship but not definitive.

Positivity

The positivity assumption requires that for every combination of covariates in the adjustment set, there is a non-zero probability of receiving both treatment levels. In this case, short and long response times. This assumption is largely plausible in this setting, as variation in EMS response time occurs across a wide range of patient profiles and operational contexts. Even if it's the same patient or location, (which we cannot know as that information is uniquely de-identified) response times and patient outcomes could likely differentiate. Violations may arise in extreme scenarios, where there may be a concentrated patient population with similar conditions/complaints (e.g. an assisted living facility), receiving the same response time. However, due to the sheer size of the dataset, it is unlikely for this to be the case without other data points upholding positivity. Use of nearest-neighbor matching also helps mitigate this by discarding unmatched units in regions of covariate space where treatment overlap is sparse, thereby improving identification at the cost of generalizability.

Consistency

The consistency assumption requires that a treatment is well-defined and that units receiving the same treatment level experience it in the same way. In this context, this means that a response time greater than the cutoff time has the same interpretation and causal effect across observations, and that there is no variation in treatment quality or exposure due to unmeasured factors. This assumption is generally reasonable in the case of response time, which is a precisely measured, protocol-driven metric. However, potential interference may exist in a system utilizing dynamic deployment of ambulances: if two neighbors call 911, one neighbor may receive their treatment faster than the other due to differing distances of the two crews. It may be assumed such effects are present, unlikely, and therefore negligible due to the size of the dataset.

Causal Estimation Strategy

To estimate the causal effect of prolonged response time on clinical worsening, we employ propensity score matching with 1:1 nearest-neighbor matching without replacement. The model includes all covariates in our DAG-derived sufficient adjustment set: $\{D, B, I\}$, as these variables are theorized to jointly influence both treatment assignment and the outcome, and are required for unbiased accurate estimation.

This approach is chosen to reduce imbalance in observed covariates between treatment groups, thereby approximating a randomized experiment and satisfying the conditional exchangeability assumption. The analysis focuses on the Average Treatment effect on the Treated (ATT), as we are primarily interested in the effect of delay among those who actually experienced it, rather than estimating a population-wide average.

We conduct a series of matches on incrementally increasing thresholds of delay (5, 9, 11, and 20 minutes). Covariates include transport urgency (lights and sirens), prior to EMS arrival bystander assistance, and delays the crews face during their response.

After matching, logistic regression is used to estimate the odds ratio (ORs) of deterioration as a function of response time threshold and other covariates. This structure allows the examination of how increasing delay thresholds affect patient outcomes in a stepped dose–response fashion.

Results

We observed a nonlinear relationship between response time and patient outcomes. At a 5 minute threshold, a longer response time than the 5 minute cutoff did not have statistical significance, (OR = 0.88, $p = 0.24$), indicating minimal clinical impact of marginally short delays. However, response times exceeding 9 minutes showed a clear inflection point where the odds of a patient's outcome worsening increased by 25%: OR 1.25 at 9 minutes ($p < 0.001$). At 11 minutes, that probability goes up to 35% (OR 1.35, $p = 0.0005$), and at 20, 65% (OR 1.65, $p = 0.009$).

This effect remained robust even after adjusting for patient age, a likely confounder in EMS outcomes, yielding a nearly identical estimate (OR = 1.26, $p = 0.0016$) at 9 minutes. The analysis also revealed an independently small but significant effect of patient age on the odds of a worsening outcome (OR = 1.005 per year, $p = 0.015$).

Bystander assistance emerged as the most consistent protective factor across all models (OR range: 0.14–0.49), associated with 51–86% lower probability of deterioration depending on the model. Lights-and-sirens responses were inconsistently significant, reflecting possible confounding, as critically ill patients are more likely to receive the expedited transport. Both bystander help and the use of lights/sirens were

significant, suggesting that early intervention and expedited transport are both independently associated with a patient's outcome. With this said, bystander intervention was the greatest protective marker.

Discussion

Overall, we see a clear correlation between rural EMS delays and worse patient outcomes. Further, this study provides robust evidence that prolonged EMS response time is causally associated with prehospital clinical deterioration, particularly when delays exceed 9 minutes. Response times at or below 5 minutes simply appear to be “fast enough,” but response times above the 9 minute mark, much less above 11 and 20 minutes, already show worsening outcomes. Although it may be intuitive to promote the increased use of lights and sirens to minimize slower response times, their use appears to have a minimal effect on preventing patient deterioration. This analysis also suggests that increased early bystander intervention can be very effective against worsening patient symptoms.

Faster response enables quicker delivery of definitive care (cpr, defibrillation, bleeding control, airway management, etc.). At the same time, not all EMS calls are equally urgent; a few extra minutes might not affect a patient with a minor injury or a stable condition. This nuance partly could explain why there is no perceived change in the probability of patient condition worsening when response times are within a lower reasonable range.

Additionally, the consistently protective effect of bystander intervention, associated with a 50–85% reduction in the odds of deterioration, highlights the importance of community training, public AED placement, and dispatcher-assisted CPR protocols. Investments in community response may buffer the harms of delayed EMS arrival.

From a policy standpoint, these findings have direct implications for response time benchmarks and performance metrics in EMS systems. While response time should not be the sole indicator of an EMS system, it remains clinically significant. In particular, maintaining response times under 9 minutes may meaningfully reduce the risk of patient deterioration in rural settings. Current national standards, often around 9 minutes for urban 911 response, cannot be universally applied in rural areas, where geography and staffing challenges are more present. However, recognizing the critical threshold may help guide triage, deployment algorithms, mutual aid agreements, and lead to alternatives to the modern EMS model.

Together, these findings suggest a dual-pronged approach: minimize harmful delays through systems-level improvements where possible, and build community capacity to stabilize patients when EMS is en route. As health systems increasingly aim to optimize

prehospital care, this work supports integrating not only temporal efficiency but also public involvement into EMS.

The rural-urban outcome gap is also influenced by differences in hospital access, baseline health status, bystander aid, and more. In urban areas, everything is closer; more bystanders likely

Limitations

Several limitations to this study and the data it uses should be acknowledged. Most critically, although we attempt to isolate a causal effect using propensity score matching and DAG-guided adjustment, our models rely on the assumption of no unmeasured confounding. While we adjust for known operational confounders including response delay, initial patient acuity, and bystander involvement, there may be other confounders not coded into the NEMSIS dataset and completely overlooked by the study. Examples could include the true severity of a patient's condition, comorbidity burden, and more. As a result, residual confounding remains a credible hazard to validity. For example, patients with underlying cardiac or respiratory conditions may be more likely to deteriorate and more likely to prompt urgent dispatch responses, biasing the treatment effect estimates upward in the absence of proper adjustment.

Second, our causal estimates are conditional on ATT, reflecting the effect of delay only among those who actually experienced it. While this design is appropriate given our goal of understanding harm among delayed patients, it does not generalize to a broader EMS population (i.e., the ATE). Our matching procedure excludes treated cases with no comparable control, particularly at extreme delay thresholds (e.g., >20 minutes), reducing generalizability and precision in these strata.

Third, although the quality and granularity of the NEMSIS dataset is excellent, there are still some practical constraints on analysis. Foremost, the dataset is a convenience sample, not a population sampling, as the data is voluntarily submitted by agencies in all 50 states. As such, there is the possibility of disproportionate data reporting in higher-resourced areas. The data is observational, too. The data cannot be considered complete and the study's conclusions aren't necessarily representative of all 911 in the nation. However, it is expected that the data is at least illustrative of the state of emergency response.

A call consists only of crew arrival and handoff to definitive care. Thus, the data can only represent the finite time and does not necessarily encapsulate a patient's true outcome at hospital discharge. Patient outcomes are also limited to pre-hospital assessments, and thus do not capture in-hospital deterioration, mortality, or long-term health outcomes, as EMS does not have the same diagnostic abilities or scope of time. A patient that was not observed to have a worsening condition for the short time in the

field may have deteriorated in the hospital during their longer stay – and it would be impossible to know if response time, hospital care, or any other factor could be a contributing factor.

Furthermore, substantial variation exists across states and agencies in terms of documentation and coding logic. While NEMSIS aims to standardize data collection, different agencies may use different charting software or have different terminology, leading to some ambiguity in the data and missing values. To mitigate this, fields were excluded with pervasive missingness and the study focused on high-quality, consistently reported and reliable variables. Nevertheless, non-random missingness and inter-agency variability could introduce measurement bias.

Within the agencies, calls are not identified as primary response or mutual aid. That is, the analysis assumes that for any given call, the same agency is sending the responding ambulance, which isn't always the case. An agency may not have any units available, and thus another agency must send their ambulance, causing significant response delay, a delay which is not accessible from the dataset.

Future Work

To strengthen causal inference and expand the scope of analysis, future research could focus on linking NEMSIS data with hospital-based records or registries to capture in-hospital outcomes and comorbidities. Such integration would enable adjustment for clinical risk factors currently unmeasured in EMS records, including underlying cardiovascular or respiratory disease, medication use, and hospital-level interventions; EMS is only a small snapshot of a patient's medical care. Additionally, machine learning methods for high-dimensional confounding control (e.g., targeted maximum likelihood estimation or causal forests) may offer promising alternatives to traditional matching, particularly as more granular and complete EMS data become available.

For lateral exploration, more research may be done on similar questions: does a shorter outcome necessarily indicate a better outcome? How might a patient's other demographics, or vitals and other call details add more granularity? How might response time and patient outcome be affected for specific types of calls, such as cardiac arrests, anaphylaxis, or even less acute calls? Finally, validation of these findings in urban and suburban settings, or in states with different EMS infrastructure, would help establish the generalizability of the identified response time thresholds.

what about shorter times means better outcomes?

how about hospital results?

add in age?

at what time cutoff does it become statistically significant?

Rural EMS simply cannot uphold the fundamental promise of EMS: emergency care at an instant. So then, what are some alternatives to the problem?

Conclusion

By leveraging a matched cohort design and adjusting for a DAG-derived confounder set, we isolated the marginal effect of delayed response while accounting for operational and situational variables such as response delay, initial acuity, and bystander involvement. The findings indicate a nonlinear dose–response relationship: delays of 5 minutes or less were not associated with harm, but at 9, 11, and especially 20 minutes, the odds of clinical worsening rose by 25%, 35%, and 65%, respectively. These results held even after controlling for age.

This study demonstrates that EMS response time is a causal and meaningful determinant of prehospital clinical deterioration, particularly when delays exceed 9 minutes. We show that delays in excess of this threshold significantly increase the odds of worsening patient acuity, while bystander assistance consistently mitigates that risk. Use of lights and sirens showed inconsistent significance, while response delays consistently increased the odds of deterioration. These findings validate some long-standing operational concerns about timeliness in EMS, but also challenge the foundations of EMS: that lights and sirens might be discouraged and education of the public should be explored more. As EMS systems adapt to geographic challenges, staffing shortages, and financial constraints, balancing speed, care, and community involvement will be central to improving patient clinical outcomes.

(Abstract II)

Background: Timeliness is a foundational component of emergency medical services (EMS), yet the causal relationship between response time and patient outcomes remains debated, particularly in rural systems where delays are common. This study evaluates whether delayed EMS response time causally contributes to prehospital clinical deterioration using national-level EMS data.

Methods: We analyzed a rural subset of the 2021 National EMS Information System (NEMSIS) dataset, focusing on 911-initiated ground responses involving patient transport. Propensity score matching was used to estimate the average treatment effect on the treated (ATT) across four response time thresholds (5, 9, 11, and 20

minutes). A directed acyclic graph (DAG) informed the identification of a sufficient adjustment set, including response delay, bystander assistance, and initial patient acuity. Logistic regression was used to estimate the causal effect of delay on a binary outcome of clinical deterioration, defined as a worsening in acuity between EMS arrival and handoff.

Results: No significant effect was observed at the 5-minute threshold. However, response times exceeding 9 minutes were associated with a 25% increase in the odds of clinical deterioration (OR = 1.25, $p < 0.001$), with effects growing stronger at 11 minutes (OR = 1.35) and 20 minutes (OR = 1.65). These results were robust to adjustment for patient age. Bystander intervention was independently associated with a 50–85% reduction in the odds of deterioration across all models.

Conclusions: Delays in EMS response exceeding 9–10 minutes causally increase the likelihood of clinical deterioration in prehospital settings. These findings provide empirical support for response time benchmarks and highlight the importance of community-based intervention strategies. This study demonstrates how causal inference methods can inform EMS performance standards and policy in rural health systems.